

Chaos in Duopoly Pricing

T.PUU

Department of Economics
University of Umeå, 90187 Umeå, Sweden

Abstract - The dynamics of two competing firms in a market is studied in terms of Cournot's duopoly theory. Assuming iso-elastic demand and constant unit production costs the iterative mapping

$$x_t = \sqrt{\frac{y_{t-1}}{a}} - y_{t-1}, \quad y_t = \sqrt{\frac{x_{t-1}}{b}} - x_{t-1}$$

for the outputs of the two firms ensues. The two constants are the unit production costs. The fixed point, the Cournot equilibrium, has earlier been assumed to be the only interesting feature of this model. It is, however, shown that the model can produce persistent motion, periodic or chaotic.

DUOPOLY

Economics recognizes two opposite market forms: competition and monopoly. In the first case the firms are very numerous and thus small in comparison to the size of the total market. In consequence they consider market price as being given independently of any action they can take alone with regard to their supply. In the second case one single firm supplies the whole market. Its supply influences market price appreciably, and it is able to increase its profits by limiting supply and establishing a monopoly price.

Planning for a monopoly is thus more complex than for a competitive firm. Oligopoly, the intermediate case between one seller and many sellers is, however, not less, but even more complex than monopoly. When there are few sellers, the oligopolists must consider the effects on market price, not only of their own actions, but of those of their competitors as well. If they can assume that the competitors take actions to retaliate upon their own previous ones, the planning situation can become extremely complex. There is a considerable existent literature especially on duopoly, the case of two sellers.

The theory of the coexistence of two sellers, each large enough to become a monopolist in the market, were it not for the competitor, dates back to Augustin Cournot in 1838 [1]. Exactly a hundred years later, in 1938, Heinrich von Stackelberg made significant additions to the theory [2]. Cournot assumed that each duopolist would just note the action of its competitor and react to it (without trying to anticipate its future reactions). The alternating adjustment process was assumed to lead to a fixed point, the Cournot equilibrium. In this article it will be shown how the simple adjustment process can instead lead to permanent

cycles and chaotic dynamics. David Rand [3] conjectured this possibility in 1978, but did not present any substantial economic assumptions that would lead to these results.

An interesting perspective is opened by von Stackelberg's concepts. The duopolist acting according to Cournot's theory is called a "follower". If one competitor adheres to this behaviour, the other can take this into consideration when calculating its best action and establish itself as a "leader", thus making a larger profit [4]. Obviously, one leader and one follower establish new possibilities. Two followers are just the Cournot case, but two leaders is an untenable situation, and it is an extremely interesting perspective to study the complex dynamics when both competitors try to become leaders. This will, however, not be discussed in the present paper.

THE COURNOT MODEL

Assume that the quantity demanded is reciprocal to price p . This represents an "iso-elastic" demand function reflecting a case where the consumers always spend a constant sum on the commodity, regardless of price. Inverting the demand function we obtain

$$p = \frac{1}{(x + y)} \quad (1)$$

where the total quantity in the denominator is the sum of the supplies, x and y , of the two competitors

The revenues of the two firms equal price times quantity, i.e., $px = x/(x + y)$ and $py = y/(x + y)$ respectively. We assume that the firms operate under constant unit costs, denoted a and b . Their total costs are accordingly ax and by . The profits become

$$\Pi_1(x, y) = \frac{x}{x + y} - ax \quad (2)$$

and

$$\Pi_2(x, y) = \frac{y}{x + y} - by \quad (3)$$

The first expression pertains to the firm having x under its control, taking y as given. In order to maximize profits for the firm we have to put the partial derivative of (2) with respect to x equal to zero, obtaining

$$x(y) = \sqrt{\frac{y}{a}} - y \quad (4)$$

In the same way we put the partial derivative of (3) with respect to y equal to zero, thus obtaining

$$y(x) = \sqrt{\frac{x}{b}} - x \quad (5)$$

A simple check of the second derivatives shows that we in fact deal with profit maxima. Equations (4)-(5) are the Cournot reaction functions, expressing the optimal behaviour of each competitor when the action of the other is taken.

To find the maximum profits we can substitute back from (4)-(5) in (2)-(3). In this way we find

$$\Pi_1^{\max}(y) = 1 - 2\sqrt{ay} - ay \quad (6)$$

and

$$\Pi_2^{\max}(x) = 1 - 2\sqrt{bx} - bx \quad (7)$$

As both expressions are perfect squares we see that profits never become negative, they just drop to zero at the respective zeros of the reaction functions (4)-(5). The advantage of the simple demand function introduced is that we do not need to be concerned about discontinuities in the reaction functions.

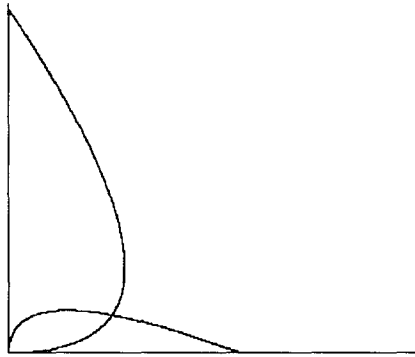


Fig. 1. Reaction functions and Cournot point.

We see the reaction functions illustrated in x, y -space in Fig. 1 along with their intersection, the Cournot equilibrium. They drop to zero at $y = 1/b$ and $x = 1/a$ respectively, and attain maxima at $y = 1/(4b) = x$ and $x = 1/(4a) = y$ respectively. The Cournot equilibrium is attained when (4)-(5) hold simultaneously. The system obtained is easily solved and yields

$$x = \frac{b}{(a+b)^2} \quad (8)$$

and

$$y = \frac{a}{(a+b)^2} \quad (9)$$

It is a quite different question whether the Cournot equilibrium would be stable in a dynamic adjustment process, even the one implicit in his own model. We suspect that this depends on the slopes of the tangents at the intersection point, and that we might easily produce cycles and chaos.

Substituting from (8)-(9) into (2)-(3) we can easily obtain the profits of the duopolists under Cournot equilibrium as

$$\Pi_1^{cnt} = \frac{b^2}{(a+b)^2} \quad (10)$$

and

$$\Pi_2^{cnt} = \frac{a^2}{(a+b)^2} \quad (11)$$

respectively. These can be compared with the profits from other kinds of equilibria, like those of the Stackelberg cases.

ADJUSTMENT BY TAKING TURNS

We now have to introduce the adjustment process explicitly. From (4) and (5) we have

$$x_t = \sqrt{\frac{y_{t-1}}{a}} - y_{t-1} \quad (12)$$

and

$$y_t = \sqrt{\frac{x_{t-1}}{b}} - x_{t-1} \quad (13)$$

The dynamics of this system will be found capable of producing persistent movement, periodic and chaotic, along with Cournot equilibria. We anticipate the discussion a little by producing Fig. 2, which shows a chaotic adjustment process in the same setting as Fig. 1. The two-dimensional mapping (12)-(13) has, however, not been repeated in the usual way. According to tradition in the economics of duopoly it is assumed that the two competitors take turns every second period each in adjusting according to (12)-(13). We will later see what difference it makes to let the firms adjust simultaneously.

To find the parameter ranges for various types of output, let us first locate the loss of stability for the fixed point. Differentiating the reaction functions (4)-(5) we obtain

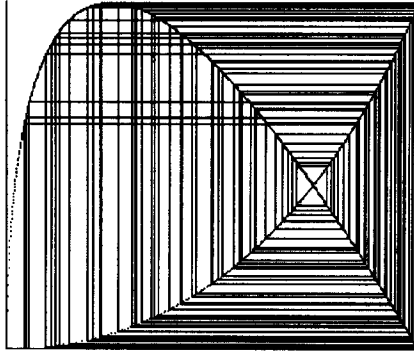


Fig. 2. Chaotic adjustment in the duopoly.

$$x'(y) = \frac{1}{2} \sqrt{\frac{1}{ay}} - 1 \quad (14)$$

and

$$y'(x) = \frac{1}{2} \sqrt{\frac{1}{bx}} - 1 \quad (15)$$

Next we substitute from (8)-(9) to obtain the derivatives in the Cournot equilibrium point

$$(x')^{cnt} = \frac{1}{2} \frac{b-a}{a} \quad (16)$$

and

$$(y')^{cnt} = \frac{1}{2} \frac{a-b}{b} \quad (17)$$

The loss of stability for the fixed point occurs when

$$(x')(y') = -1 \quad (18)$$

Substituting from (16)-(17) into (18) we get the condition in the form

$$(a-b)^2 = 4ab \quad (19)$$

This can be solved for a/b or b/a equal to $3 \pm 2\sqrt{2}$. We recall that this is the ratio of the unit costs of the two firms. Whenever this ratio is within the interval $(3 - 2\sqrt{2}, 3 + 2\sqrt{2})$ then the fixed point is stable.

Whenever the ratio falls outside this interval we at least have period doubling. For interesting dynamics the unit price ratio must be relatively far from unity, in terms of economic substance we have to consider the case of one enterprise with high overhead but low unit costs, and another with low overhead but high unit costs, like one old and one modern firm.

Like other similar models this one is apt to explode for certain parameter values. We already found that the reaction function (4) has a zero for $y = 1/\alpha$ and that it has a maximum for $y = 1/(4\alpha)$. Its corresponding maximum value is $x = 1/(4\alpha)$ as well. The reaction of the other firm on this maximum value can be obtained by substituting this into (5) as $y = 1/\sqrt{4\alpha b} - 1/(4\alpha)$ and this value must not overshoot the zero of (4), or the model will explode. The condition for non-explosion reads

$$\sqrt{\frac{1}{4\alpha b}} < \frac{5}{4\alpha} \quad (20)$$

or

$$\frac{a}{b} < \frac{25}{4} \quad (21)$$

This reasoning could also be carried out starting out with the other reaction function, so that we could establish the condition

$$\frac{a}{b} > \frac{4}{25} \quad (22)$$

We conclude that both the emergence of period doubling and explosion of the model depend on the ratio of marginal costs. As mentioned earlier it has to fall outside the interval $(3 - 2\sqrt{2}, 3 + 2\sqrt{2})$ to produce something more than just fixed points, and we saw that it has to fall inside the interval $(4/25, 25/4)$ in order not to blow up the model. The parameter intervals for interesting dynamics, periodicity and chaos, are indeed tiny, but we should not be too much concerned about that. The demand function and the cost functions were indeed very special in terms of their simplicity, though quite reasonable from the viewpoint of general economic theory. We should thus be content with the mere existence of these bands of periodicity and chaos, trusting that they might be considerably extended under slightly different assumptions.

In Fig. 3 we show the amplitude response of commodity price over a range of unit price ratios. We note that there is a period doubling route to chaos. As mentioned this represents a small section of the parameter interval, in the dominant middle section we just have a fixed point and nothing else.

SIMULTANEOUS ADJUSTMENT

Let us now assume that the competitors adjust simultaneously in stead of taking turns. We still use the system (4)-(5), but now interpret it in the usual way. The result of the process for the same parameters as

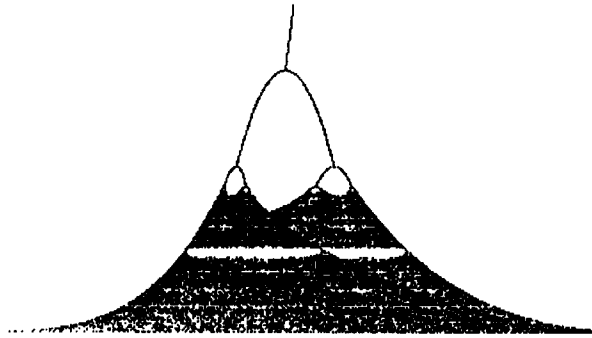


Fig. 3. Section of chaotic, periodic, and equilibrium response to the unit cost ratio

illustrated in Fig. 2 now looks like the picture in Fig. 4.

We note the interesting feature that the attractor consists of sets of vertical and horizontal lines so spaced that they intersect on the adjustment curves. The whole set could be traced by following a spiral outward motion outwards through the nested rectangles. We also note that there is a blank never visited cross through the Cournot equilibrium. Nothing essentially new is, however, attained by the simultaneous interpretation of the mapping, as the bounds for the stability of the model are still those already located.

The situation, however, becomes different once we let both variables be present in the right hand sides of the reaction functions (4)-(5). The easiest and most reasonable way to accomplish this is to assume an adaptive process by which the competitors do not immediately jump to their new optimum in each step, but adjust their previous decision in the direction of the new optimum. Thus

$$x_t = x_{t-1} + \theta \left(\sqrt{\frac{y_{t-1}}{\alpha}} - y_{t-1} - x_{t-1} \right) \quad (23)$$

and

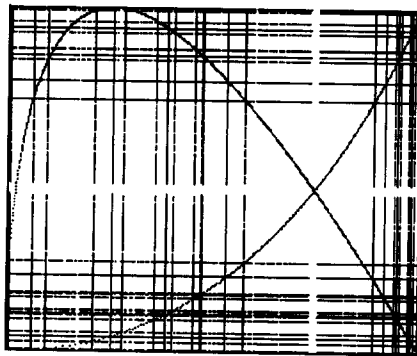


Fig. 4. Chaos in simultaneous adjustment.

$$y_t = y_{t-1} + \theta \left(\sqrt{\frac{x_{t-1}}{b}} - x_{t-1} - y_{t-1} \right) \quad (24)$$

where $0 \leq \theta \leq 1$. If $\theta = 1$ we deal with the case already discussed. The system in general can now no longer be written as a pair of recursive equations for each variable separately, and we expect quite different types of attractors in phase space, like the one in Fig.5.

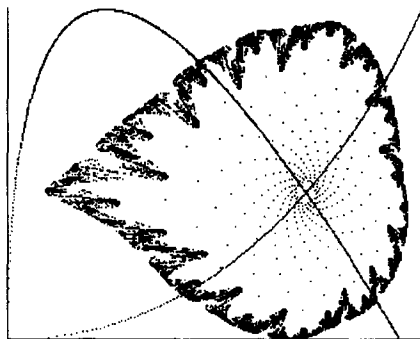


Fig. 5. Chaotic attractor of the adaptive system.

The unit cost parameters are close to those used in the previous pictures and the adaptation coefficient is rather high, 0.9. In Fig. 5 we see how the transient process diverges from the neighbourhood of the unstable Cournot equilibrium and is attracted to the chaotic "leaf". For other parameter values the attractor becomes a closed curve. In general the adaptivity assumption serves to stabilize the model.¹

CONCLUSION

It is hardly surprising that Cournot's duopoly theory is capable of producing chaotic dynamics, provided suitable non-linearities are incorporated into the model. As mentioned, this was anticipated by Rand [3]. The remaining problem was to introduce the non-linearities in terms of reasonable principles from basic economics. There certainly exist such principles. The production costs could typically be represented by a cubic polynomial, and the demand function could be quite complex as well, provided there are different segments of the market with different demand elasticities. It is, however, not easy to derive useful reaction functions from such principles, and therefore the analysis starts from the reaction functions themselves. This has made it difficult to interpret the economic substance of the models suggested.

The advantage of the present model is that it starts from so simple assumptions that the reaction functions can be explicitly derived, and yet is capable of producing complex dynamics.

REFERENCES

1. A. Cournot, *Récherches sur les principes mathématiques de la théorie des richesses*. Paris (1838).
2. H. von Stackelberg, Probleme der unvollkommenen Konkurrenz, *Weltwirtschaftliches Archiv* 48:95 (1938)

3. D. Rand, Exotic phenomena in games and duopoly models, *Journal of Mathematical Economics* 5:173 (1978)
4. J.H. Henderson and R.E. Quandt, *Microeconomic Theory - A Mathematical Approach*. McGraw-Hill, New York (1958)